

JOSHUA C. HU

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EDUCATION

University at Buffalo

B.S. in Computer Science, Concentration: Intelligence, Modeling & Simulation

Buffalo, NY

Expected May 2027

University of Toronto

Bioinformatics & Data Science

Toronto, ON

Sep 2023 – May 2024

EXPERIENCE

AI Fellow

May 2026 – Apr 2027

Break Through Tech, Cornell University

Remote

- Selected for a year-long AI Fellowship with **Cornell Tech**: intensive summer ML Foundations training plus a fall-spring team industry challenge project with a partner company such as **Google**, **Snowflake**, or **KPMG**.

LLM Researcher Intern

May 2026 – Present

Algoverse

Remote

- Built an **LLM evaluation framework** for cost-regularized agentic RL (GRPO): a gated **6-experiment** pipeline with oracle-validated measurement, Bonferroni-corrected statistical gates, and multi-seed reproducibility, plus an online **guardrail** that eases the cost penalty from live training signals to catch policy collapse early (paper under review).

Software Engineering Intern

May 2026 – Aug 2026

Linde

Buffalo, NY

- Redirected the reporting platform to Microsoft Fabric/Power BI after evaluating **4** architectures, cutting **~60–70%** of planned custom development and positioning the platform for LLM-driven reporting.
- Designed a strangler-fig migration from an aging on-prem SQL row-store to hybrid KQL/Eventhouse + SQL on Microsoft Fabric for **30,000+** IoT devices on a **3.17B-row** telemetry workload; columnar ingest eliminates burst lock contention during mass RTU reconnection and delivers **5–8× storage compression**.

Software Tooling Developer

Jan 2026 – Present

University at Buffalo, CSE Department

Buffalo, NY

- Refactoring a multi-institution code-tracing platform under guidance of a **Notion** engineer; architecting a React + TypeScript + PostgreSQL stack with a Docker deployment pipeline for cross-university adoption.
- Scaling the platform from **800+ students** and **15,000+ submissions** to multi-institution use, with one external university in early evaluation for a potential Fall 2026 rollout and system outputs positioned for academic publication.

AWS AI & ML Scholars — Agent Developer Track

Aug 2025 – Nov 2025

Amazon Web Services

Remote

- Selected for the fully-funded program as a **top ~3%** performer (**top 1,500 of 50,000+**) based on performance.
- Built a health-claims triage agent on **Amazon Bedrock** with an **S3 + Lambda** ingestion pipeline, FAISS vector retrieval, and **guardrails** that raised the compliance-check pass rate **~25%** over an unguarded baseline.

Teaching Assistant

Aug 2025 – Present

University at Buffalo

Buffalo, NY

- Supported **453+ students** across labs, office hours, and grading; ran structured debugging workshops that lifted mentee exam scores **~20% above class median**, and built tailored materials for non-native speakers and students with disabilities.

LLM Researcher Intern

May 2025 – Sep 2025

Algoverse | Columbia, Meta FAIR, UofT, St. Jude Children's Research Hospital

Remote

- Co-authored a **NeurIPS 2025 Workshop** paper (LoRA-Guided PPO, second author) on **cost-aware agent orchestration**; built a controller that decides which sub-agents to invoke under a budget, cutting **inference cost-per-success 12%** vs. a PPO baseline (**24.30 → 21.40**) while holding **100% task success** on ToolBench.
- Engineered a DAG-based multi-agent orchestrator with on-demand subagents and vector-DB memory; parallel RAG workflows handled tasks **3× larger** than a single context window while cutting **token cost 28%**.

PROJECTS

StructStudio | React, JavaScript, Canvas API, Google Analytics, Playwright

Feb 2026 – Present

- Co-founded an interactive code-visualization platform **with one other engineer** as an inaugural UB Forge fellow (**1 of 14** selected from **250+** applicants); renders real-time simulations of Trees, Heaps, and Graphs from raw program input, shipped to production and adopted across theory-level CS courses at the University at Buffalo, including CSE 250 (Data Structures & Algorithms).
- Engineered a React rendering pipeline achieving **sub-100ms render times** while sustaining **300+ concurrent users**; instrumented with Google Analytics for usage telemetry and Playwright for end-to-end test coverage.

LEADERSHIP

Data Science & Machine Learning Club — Founder & President

Jun 2025 – Present

University at Buffalo

Buffalo, NY

- Founded and scaled the club to **120+ members**; secured partnerships with the UB AI/ML department to source speakers from **Amazon**, **Google**, and **Tesla**, and built real-world project briefs for **6+ technical workshops**.
- Managed a **6-person executive team** and **\$1.5K+ annual budget**, structuring operations across event logistics, outreach, and content planning to sustain weekly programming.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, KQL, C, HTML, CSS

AI / ML: PyTorch, TensorFlow, scikit-learn, LangChain, FAISS, pandas, NumPy, Embeddings, RAG, LoRA, PPO, GRPO

Web & Frameworks: React.js, Next.js, Node.js, Express.js, REST APIs, WebSockets

Tools & Platforms: Git, Docker, AWS (S3, EC2, Lambda, SageMaker, Bedrock), PostgreSQL, Vercel, Microsoft Fabric, Claude Code

Concepts: Multi-Agent Systems, Agent Orchestration, Evaluation Frameworks, AI Guardrails, System Design, Vector Databases